

A novel molecular method for noninvasive sex identification of order Carnivora

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Abstract Noninvasive sampling can provide an efficient means of genetically monitoring mammals. Due to the fragmented quality of DNA derived from such samples, few methods span taxonomic groups beyond species. Here, we describe a more universal protocol for the molecular sex identification of order Carnivora. PCR amplification of a 176 bp segment of the *SRY* gene indicates a male, whereas no amplification indicates a female. A 245 bp segment of the *ZFX* gene serves as a positive control. This method is internally consistent, and multiple trials confer confidence in results. It applies to at least three families (Canidae, Felidae, and Phocidae) in both carnivore suborders (Feliformia and Caniformia), rendering this protocol the first sex identification method with amplicon sizes appropriate for noninvasive samples suitable for multiple families within order Carnivora.

Keywords Noninvasive DNA sampling · Sex assignment · Carnivora · *SRY* · Canidae · Felidae

Noninvasive sampling can be ideal for monitoring mammalian carnivores. Through reliance on hair or excreta, noninvasive sampling eliminates the need to locate elusive species and handle endangered organisms (Waits and Paetkau 2005).

Molecular sex identification is one application of non-invasive sampling. By amplifying segments on the X and Y chromosomes, these methods enable sex ratio determination for analyses on behavior, demography, and population viability. Fragmented DNA limits most noninvasive methods to species (e.g., Trinca and Eizirik 2012) due to the difficulty of designing widely applicable primers with small amplicons. Few exceptions, including Pilgrim et al. (2005), Wei et al. (2007), Pagès et al. (2009), and González et al. (2015), target multiple species within a family with small amplicon sizes between 144 and 440 bp.

The current study expands upon these efforts by targeting a broader taxonomic group. Through novel primer design and PCR optimization, we developed a reliable method for sex identification of multiple families within order Carnivora with small amplicon sizes (176 and 245 bp) appropriate for use with noninvasive samples.

We designed primers for the sex-determining region (*SRY*) of the Y chromosome (Table 1) by identifying highly conserved regions across Felidae in Geneious version 6.1.7 (<http://www.geneious.com>, Kearse et al. 2012). We included multiple genera to increase the likelihood of broad applicability, but restricted the search to felid sequences (DQ095161, DQ095163, DQ095165, DQ095177, DQ095178, DQ095180, DQ095184, DQ095188, DQ095190, DQ095193) to maximize candidate segments analyzed.

To identify a positive control, we searched relevant literature for loci on the X chromosome. We selected a zinc-finger (*ZFX*) primer pair (Table 1) designed by Murphy et al. (2001) for its applicability to placental mammals, melting temperatures near those of our *SRY* primers, and small amplicon.

We optimized our method with DNA extracted from control individuals of known sex. DNA was not quantified but was assumed to vary in concentration and quality as is

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Table 1 A novel *SRY* primer pair and the *ZFX* control designed by Murphy et al. (2001)

Locus	Primer sequence 5'–3'	T _m (°C)	Expected ^a /observed ^b size (bp)
<i>SRY</i>	CSRY1F: CGCAAGGTGGCTCTAGAG ^c	55.3	180/176
	CSRY1R: GCCTTSCGACGAGGTCGGTA	61.6	
<i>ZFX</i>	ZFX1F: TGGCCCAGATGGACATCC ^c	57.3	242/245
	ZFX1R: AATGCTTTCCGGACTCATCG	55.3	

^a Calculated from GenBank sequences^b Observed during GeneMapper trials^c Fluorescently tagged

typical of genomic DNA extracted from scat. Samples included tissue from 72 Mediterranean monk seals (*Monachus monachus*) and 10 Hawaiian monk seals (*M. schauinslandi*), blood from 48 pumas (*Puma concolor*), and scat from 25 Pallas's cats (*Otocolobus manul*), 4 jaguars (*Panthera onca*), 2 tigers (*P. tigris*), 1 lion (*P. leo*), 1 serval (*Leptailurus serval*), 2 bobcats (*Lynx rufus*), 2 wolves (*Canis lupus*), 1 domestic dog (*C. l. familiaris*), and 2 coyotes (*C. latrans*). Blood and tissue DNA extractions were performed using the DNeasy Blood and Tissue Kit (Qiagen Inc.). Scat extractions followed a modified QIAamp DNA Stool Mini Kit (Qiagen Inc.) protocol (Caragiulo et al. 2014).

Multiplex polymerase chain reaction (PCR) was conducted in 25 µL reactions containing 12.5 µL Multiplex PCR Master Mix (Qiagen, Inc.), 1 µL of each primer (10 µM), 6 µL water, and 2.5 µL template DNA. For scat samples, we added 0.1–0.3 µL BSA (Roche, 1 mg/mL) to increase the likelihood of amplification.

Touchup PCR conditions consisted of initial denaturation 94 °C/15 min followed by [94 °C/30 s, 49 °C/30 s, 72 °C/45 s] × 5 cycles, [94 °C/30 s, 51 °C/30 s, 72 °C/45 s] × 15 cycles, [94 °C/30 s, 53 °C/30 s, 72 °C/45 s] × 20 cycles, [94 °C/30 s, 55 °C/30 s, 72 °C/45 s] × 15 cycles, and final extension 72 °C/10 min. For high quality samples, we decreased the number of cycles to 15 at 53 °C and 10 at 55 °C.

We analyzed PCR product through agarose gel electrophoresis and peak analysis in GeneMapper. Amplification of the 176 bp *SRY* segment and 245 bp *ZFX* control indicated a male. If DNA was present with high enough quality to amplify the larger segment, failure to amplify *SRY* suggested that no Y chromosome was present in the sample, indicating a female. For best results, we performed three trials per sample and analyzed peaks in GeneMapper.

This method has proven effective for species in Felidae, Phocidae, and Canidae. To assess applicability to Mustelidae, Odobenidae, Otariidae, and Ursidae, we ran in silico PCRs using Primer3 in Geneious. Successful trials included sea otter (*Enhydra lutris*; AY424668), Eurasian badger (*Meles meles*; AB539136), walrus (*Odobenus rosmarus*;

AY424653), Antarctic fur seal (*Arctocephalus gazella*; AY424651), Steller sea lion (*Eumetopias jubatus*; AY424649), giant panda (*Ailuropoda melanoleuca*; AM748312), Malayan sun bear (*Helarctos malayanus*; AB292067), and American black bear (*Ursus americanus*; AM748307) sequences. Up to two mismatches occurred in the reverse primer, but successful seal trials showed that the method still works under proper wet lab conditions.

When repeated in triplicate on 72 Mediterranean monk seal tissue samples, sex assignments matched the observed sex of every individual amplified except one. Furthermore, results were internally consistent. 86.1 % of samples yielded the same amplification patterns in all three trials. The remaining 13.9 % yielded two consistent results (i.e., the number of trials needed to assign sex).

Such broad applicability, accuracy, and internal consistency render this protocol the first sex identification method with amplicon sizes appropriate for noninvasive samples suitable for multiple families within order Carnivora. Its adoption can enable more efficient monitoring, management, and conservation of carnivores with minimal organismal disturbance.

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